

Electrical Characterization of Indoor Air Quality in the Presence of Various Natural Air Purifiers

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ABSTRACT: Air pollution is a critical threat to human health and the quality of life in large cities. In this work, we electrically characterized indoor air quality in Dhaka City with a microcontroller-based advanced sensing system in the presence of 60 air purifiers. We conducted LabVIEW-controlled, fully automated, and remotely operated experiments to precisely monitor, store, and analyze the air-purifying effects in the concentrations of air quality index (AQI) parameters—PM_{2.5}, CO₂, CO, HCHO, and TVOC. These stimulants were released into the air as vapor diffused from 32 essential oils from an aroma diffuser and 27 boiled herbs/spices from a water heater. From these 59 natural substances, the stimulant concentrations were obtained using calibrated sensors and a commercial meter, and the results are referenced with those from a commercial air purifier. The best PM_{2.5}, CO₂, and CO suppression was observed with the boiled black tea, fennel seed, and citrus peel vapor, while effective HCHO and TVOC reduction was obtained with the vapor from boiled thyme and essential oil of ginger. Decent reductions of CO₂ and CO resulted from the essential oils of lemongrass, jasmine, saffron, and strawberry. The research outcomes of this work can add great medicinal benefits and healthy alternatives to treat patients with respiratory ailments and improve the overall well-being of city dwellers.

1. INTRODUCTION

Air pollution is a lethal adversary to human health, being the fourth highest risk yet the most ignored one.¹ The health adversities span a plethora of diseases and disorders, the most common ones being related to stroke, pulmonary complexities, respiratory issues, neurological disorders, and numerous other fetal health disorders.² Worldwide, air pollution kills 7 million lives, with a significant group belonging to middle- and low-income countries.³ While the exterior sources of pollutants are industries, vehicle emissions, air conditioners, construction sites, and so on, the interior pollutants include emissions from infiltration from outside, cooking stoves, household appliances, etc.

From the stated death rate above, the ambient air pollution death toll is 4.2 million; indoor pollution kills 3.8 million, and 1 million die due to both factors.³ Dhaka, the capital of Bangladesh, stands 24th in the list of most polluted cities in the category of air, and for decades, the consistency has been maintained.⁴ The rankings are based on the average air quality index (AQI) over a specific time frame. The contributing factors

comprise unplanned urbanization, lax transport regulation, improper monitoring of industry emissions, poorly ventilated buildings, and a lack of awareness by a significant fraction of city dwellers.

While overall air quality improvement is undoubtedly a critical concern, addressing this issue in densely populated urban areas poses significant challenges. Nevertheless, improving indoor air quality at individual levels is within one's control. In addition, comparatively, much of our time is spent indoors, whether at home, school, work, or recreation spaces.

Moreover, indoor air quality is often inferior due to air stagnation. Commercial air purifiers could be a beneficial

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alternative. Nonetheless, those are out of the affordability reach of the majority of the population. A good range of indoor plants also mitigates detrimental pollutants in indoor air. However, the plants require nurturing and pest control. According to a study by Cummings et al., the ratio of plant quantity to room size to absorb the harmful elements of indoor air is gigantic.⁵ Not only does the presence of a large number of plants in a room hinder the functionality of the room but also it involves a substantial expense. Other advantageous substitutes could be diffused essential oils and water-soaked boiled herbs, which require minimum space, attention, and effort. Essential oils have proven to comprise innumerable benefits in culinary applications, health, cognition, beauty, and air-cleansing by inhibiting microbial activities, pest control, etc.⁶ Figure 1 illustrates the reported health benefits of numerous well-known essential oils. However, their efficacy in alleviating the AQI value is debatable due to the lack of adequate precisely measured data and conclusive studies.

In our study, the effects of diffused essential oils and water-soaked boiled herbs on the AQI parameters were quantitatively measured and analyzed by using a LabVIEW-controlled, fully automated unmanned setup. Several AQI parameters were recorded, such as particulate matter 2.5 (PM_{2.5}), carbon dioxide (CO₂), carbon monoxide (CO), total volatile organic compound (TVOC), and formaldehyde (HCHO) concentrations. We found that the vapors from a good number of essential oils and boiled herbs can successfully push down the AQI parameter levels in the air. A rigorous comparative study is presented in this paper on 60 air cleansers based on the percentage changes in the concentrations of the AQI parameters over 3 h of operation.

2. LITERATURE REVIEW

AQI is an indicator of air cleanliness. Generally, it comprises 5 major constituents: oxides of carbon, oxides of sulfur, oxides of nitrogen, ground-level ozone, and particulate matter (PM_{2.5} and PM₁₀). A specific range of the density of gases corresponds to a particular band of AQI for each component. However, the relationship is not linear. The correspondence of the gas density range and the AQI bands varies from region to region based on geographical and situational conditions.

At a specific time, the component resulting in the highest AQI value, usually PM_{2.5} or PM₁₀, dominates the overall AQI value. Although all of the constituents are hazardous in their own accord, PM is the most prominent. PM comprises the nongaseous air particles confined in water droplets that stay afloat in the air. Once inhaled, the PM encroaches throughout the bloodstream, affecting the organs on its way and damaging the nerves.⁸ The AQI band values, health alerts, and equivalent PM_{2.5} values are given in Figure 2 and Table 1. In Table 1, we can observe that PM₁₀ and AQI have quite similar band values with slight deviation; for example, PM₁₀ of a band value 55 to 154 links to an AQI band value of 51 to 100.

Equation 1 calculates the AQI parameter level, and the descriptions of the contributing variables are listed in Table 2.

$$I_p = \frac{I_{Hi} - I_{Lo}}{BP_{Hi} - BP_{Lo}}(C_p - BP_{Lo}) + I_{Lo} \quad (1)$$

Su et al. demonstrated the effects of diffusing essential oil on the levels of CO, CO₂, TVOC, PM₁₀, and microbes. The 3 most common oils, lavender, eucalyptus, and tea-tree oils, were able to reduce microbe activity for a good period while elevating

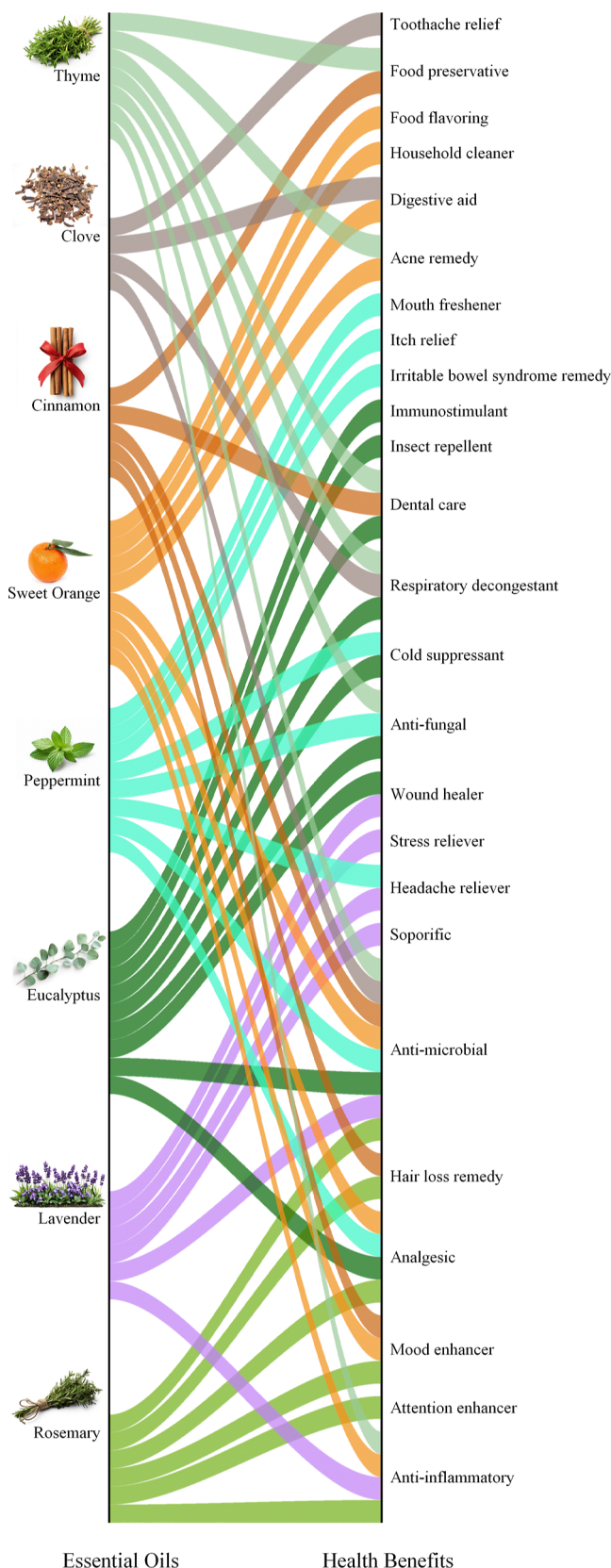


Figure 1. Commonly known health benefits of essential oils. Images of herbs and spices are taken from various free-domain websites without copyright restrictions.

TVOC, CO₂, and CO, leaving PM₁₀ nearly constant. The rise of the constituent levels was basically due to the incensing of the

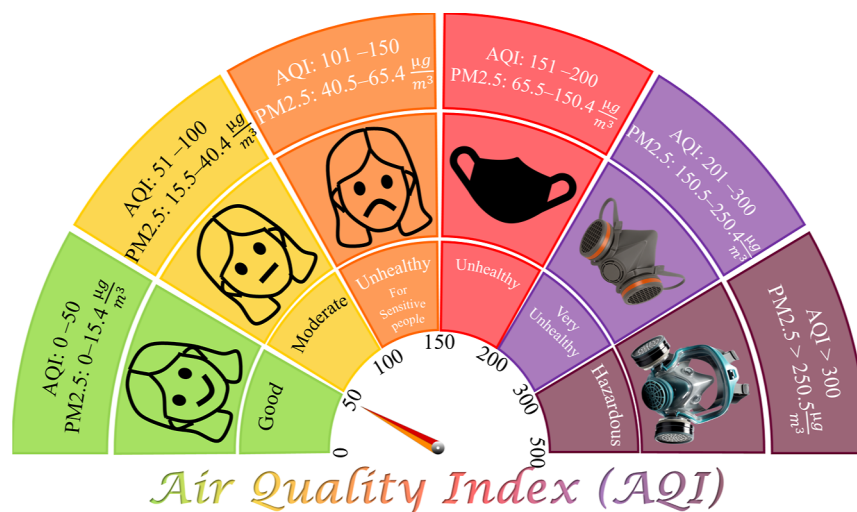


Figure 2. AQI band with representative color and corresponding health alert. With increasing AQI levels, the health hazard intensifies. Images and icons of faces are taken from various free-domain websites without copyright restrictions.

Table 1. AQI Parameter Band Range and Corresponding Health Alert⁷

AQI	PM10	PM2.5	CO	health alert
0–50	0–54	0.0–15.4	0.0–4.4	good
51–100	55–154	15.5–40.4	4.5–9.4	moderate
101–150	155–254	40.5–65.4	9.5–12.4	unhealthy for sensitive groups
151–200	255–354	65.5–150.4	12.5–15.4	unhealthy
201–300	355–424	150.5–250.4	15.5–30.4	very unhealthy
301–400	425–504	250.5–350.4	30.5–40.4	hazardous
401–500	505–604	350.5–500.4	40.5–50.4	very hazardous

Table 2. Description of AQI Terminologies

variable name	description
I_p	the index of a particular pollutant
C_p	the concentration of the pollutant
BP_{Hi}	the breakpoint that is greater than or equal to C_p
I_{Hi}	the corresponding AQI value of BP_{Hi}
BP_{Lo}	the breakpoint that is less than or equal to C_p
I_{Lo}	the corresponding AQI value of BP_{Lo}

essential oil with a candle. In addition, the experiments were performed in a ventilated room, exposing the measurements to external factors. Moreover, the experiments were not unmanned, comprised only three essential oils, and had a low data acquisition rate of 1 min per sample.⁹

Our study involved the use of an electrical diffuser and water heater to generate vapor from natural air purifiers, including 32 essential oils and 27 water-soaked herbs or spices, eliminating the requirement of oil heating or burning. Our exhaustive list of 59 natural air purifiers includes commonly known herbs and several South Asian herbs reputed for medicinal use. In addition, the experiments were carried out within a narrow outdoor AQI range within a season to maintain both short-term and long-term ambient uniformity. The experiments were fully automated and controlled by LabVIEW, thus requiring no human presence in the experiment room. Moreover, the data were collected for a 3 h duration with a high data acquisition frequency of a minimum of 2 s per sample. Additionally, 2 sets of sensors were involved to confirm the measurement accuracy of each characterized AQI parameter.

3. EXPERIMENTAL SETUP AND METHODOLOGY

Our experiments involved a combination of numerous hardware and software tools. To fully automate the system, we used microcontrollers, cameras, and actuators connected to a computer running MATLAB, LabVIEW, OBS Studio recorder, and a virtual serial port emulator.¹⁰

Additionally, we used a commercial aroma diffuser and an electric water heater to diffuse the ingredients to be tested. Figure 3a shows a block diagram of the entire experimental setup. In the following sections, we discuss the software and hardware tools used to run the system successfully for a long period.

3.1. MATLAB. We utilized the image processing toolbox of MATLAB to perform the optical character recognition (OCR)^{11,12} to capture images of the DM72B Air Quality Monitor screen using a camera (Rapoo C280). The meter screen shows PM2.5, HCHO, TVOC, and CO₂ information.

For the OCR, we adjusted the camera properties using a video capturing tool (OBS Studio), as shown in Figure 3b. The topmost picture of Figure 3b shows the live video of the meter display screen captured by the camera, and the bottom one shows the OBS-adjusted image fed to MATLAB. OBS adjusted the contrast, saturation, and brightness of the image, yielding significantly reduced OCR errors. Figure S1 of Supporting Information Note S1 shows the camera filter adjustment settings using the OBS software (Figure S1).

MATLAB code was used to create a variable to store information from the camera via the USB hub. Then, the picture was previewed, and a region of interest (ROI) was manually marked, containing only the information on the required data. This step was followed by MATLAB configuring a virtual serial port with a baud rate of 9600 to send data to LabVIEW. While running the recognition process, the code also checked whether meaningful data was received for a specific AQI parameter. Additionally, the data blocks were concatenated and structured as a single data string. Finally, the data string was sent to LabVIEW via the virtual serial port and displayed on the screen. Moreover, LabVIEW stored the data in an Excel file. This cycle continued until the terminal condition was met.

3.2. Sensors. For our research, we experimented with several well-known sensors for each AQI parameter under the same environmental conditions to find the best sensor. The

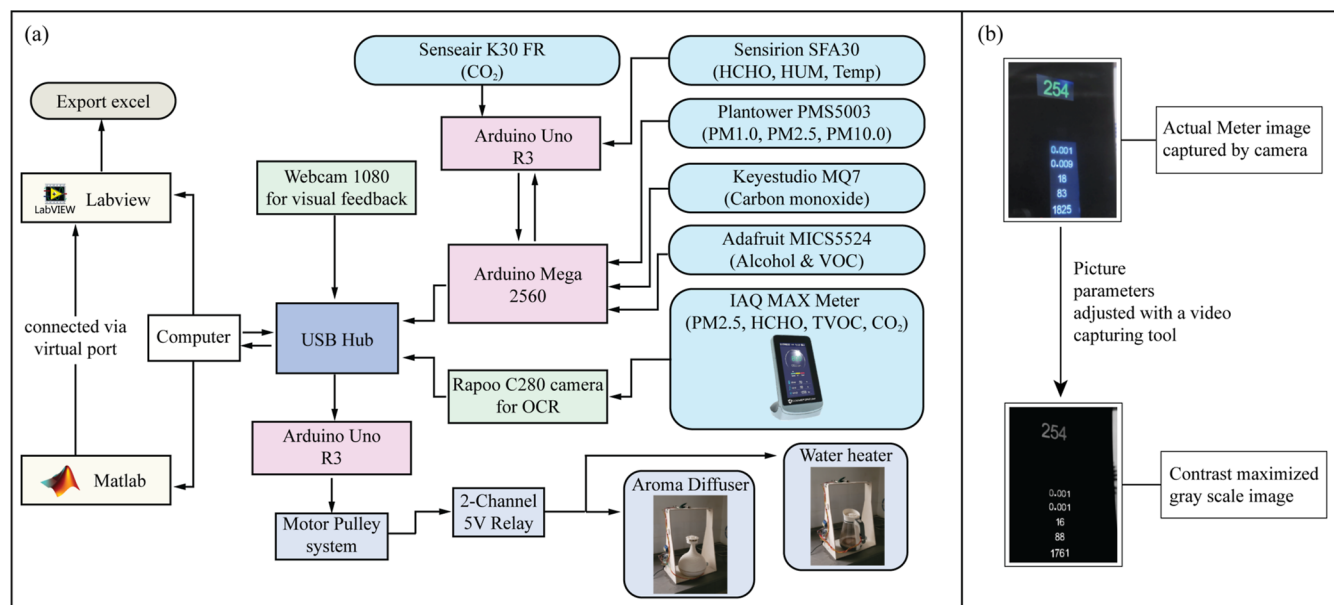


Figure 3. (a) Block diagram of the entire experimental setup. (b) Meter display image filtered using OCR with a camera.

Table 3. Sensor Parameters and Properties

*model	Senseair k30 FR	Sensirion SFA30	Plantower PMS5003	Keystudio MQ7	Adafruit MiCS5524
measurement	CO ₂	HCHO, hum, temp	PM1.0, PM2.5, PM10.0	CO	CO, ethanol, hydrogen, ammonia, methane
range	0–5000 ppm	<1000 ppb, 10–90% RH, <40 °C	<1000 µg/m ³	300–10,000 ppm	<1000, 10–500, <1000, <500 and >1000 ppm
accuracy	±30 ppm	±30 ppb, ±5%	0.3 µg/m ³		
communication	UART	I ² C	UART	analog voltage	analog voltage
response time	2 s	<2 min	<1 s	<1 s	<1 s
operation principle	nondispersive infrared	electromechanical and RHT sensor	laser scattering principle	Al ₂ O ₃ ceramic tube & SnO ₂ layer	MOS sensor

parameters of these sensors are listed in Table 3. We tested various characteristics, and further details are given below:

- A carbon dioxide sensor by Senseair was used to measure the concentration of CO₂ in the air. This sensor uses nondispersive infrared (NDIR)¹³ to detect CO₂.
- The particulate matter sensor by Plantower gave information about PM in the air, specifically PM1.0, PM2.5, and PM10.0. However, for our experiment, we were only interested in PM2.5.
- The MQ7 sensor by Keystudio was used to measure the carbon monoxide content. It uses gas-sensitive material SnO₂,¹⁴ which has lower electrical conductivity in clean air. The electrical conductivity of the sensor increases with the rise of the carbon monoxide concentration in the air.
- To measure the concentration of volatile organic compounds (VOC) in air, we used a compact metal oxide sensor (MOS)¹⁵ by SGX Sensortech. We put it together on a PCB board by Adafruit Industries (Adafruit MiCS5524).
- The characterization system also contained another commercially available AQI meter for cross-checking the collected data from sensors. DM72B Air Quality Monitor measured parameters like carbon dioxide (400–5000 ppm) and PM2.5 (0–999 µg/m³), HCHO (0–1.999 mg/m³), TVOC (0–9.999 mg/m³), temperature (–10 to 50 °C), and relative humidity (20–85% RH).

- Sensirion SFA30 gave information on formaldehyde using its electromechanical sensor with humidity and temperature by its RHT (Relative Humidity and Temperature) sensor.

A list of the functional properties of the sensors used in this work is shown in Table 3.

3.3. Microcontroller. We selected Arduino boards as our microcontrollers to keep our setup simple yet reliable. Figure 3 shows a connection diagram of the microcontrollers. Mainly, 3 boards of Arduino were used to collect the data from sensors and control the placement of components inside the room.

All the sensors were connected to a master microcontroller (Arduino Mega) and a slave microcontroller (Arduino UNO).¹⁶ The Arduino Mega 2560 was our optimal choice because of its four hardware serial ports (we utilized three ports), adequate analog and digital pins, 256 kilobytes of storage to store the program, and a processor speed of 16 MHz.¹⁷ The Plantower PMS5003 sensor (PM sensor) sent data to an Arduino Mega through Serial communication. The Keystudio MQ7 (CO sensor) and Adafruit MiCS5524 (VOC sensor) sent data through 10 bit resolution analogue ports of Arduino Mega.

On the other side, Senseair K30 FR (CO₂ sensor) and Sensirion SFA30 (HCHO, humidity and temp sensor) sent data to the Arduino Uno via software serial communication. The Arduino Uno was again connected to the Mega through the hardware serial connection.



Figure 4. (a) Water heater setup for ingredients to be dissolved in water. (b) Aroma diffuser setup for ingredients like essential oils. (c) Sensors and microcontroller setup. (d) Photograph of the experimental setup. (e) Frontend view of LabVIEW.

The Arduino Mega received all the information, checked if all of the data were correctly received, and created a string to send to LabVIEW via its third hardware serial port. The KiCad circuit diagram of the entire system is illustrated in Supporting Information Figure S2 and the backend of the LabVIEW program is illustrated in Figure S3.

3.4. LabVIEW and Automation Procedure. We used LabVIEW to automate the whole measurement system.¹⁸ Each experiment was 4.5 h long. After the warm-up period of an initial 1 h and 30 min, data acquisition on LabVIEW started by creating an Excel file.

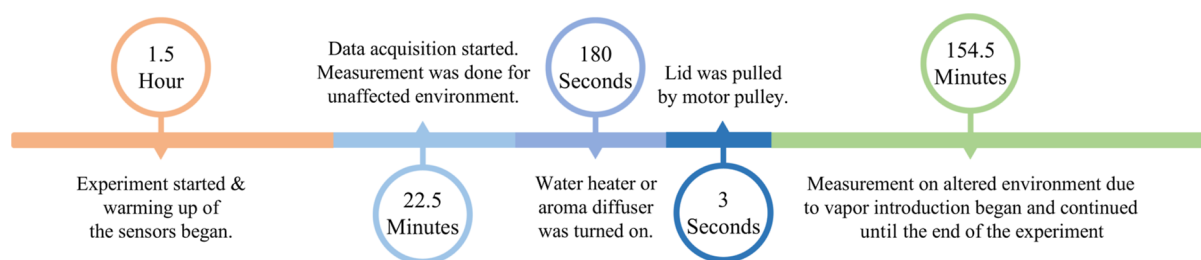


Figure 5. Measurement sequence and timeline diagram of a single experiment. The adjacent timing indicates the duration of each step. Not drawn to the scale.

The warm-up time was crucial for the sensors to ensure maximum accuracy as the temperature of gas sensors influences their sensitivity, selectivity, and response time.¹⁹ For the next 22.5 min, the sensors collected the original AQI parameters of the setup room. It gathered unmodified environmental data from all of the sensors and meters during this time.

The setup room was 923.5 ft³ (26.15 m³) in volume. We thoroughly ventilated the room with a decent-sized window, a door, and an exhaust fan. During the experiment, however, we kept the room closed to ensure an undisturbed and uniform environment throughout the experiments. A new sample (ingredients, i.e., essential oil/boiled herb) was automatically introduced inside the room. LabVIEW continued to collect the modified AQI values for the next 157.5 min with a data acquisition rate of 2 s per sample.

We aimed to make the system completely disturbance-free, ensure the same duration for all of the experiments, and eliminate human interventions. Figure 4a shows the setup of the water heater. At the end of the warm-up period, the water heater, containing the air-cleansing substance soaked in water, automatically turned on and remained on for 180 s (boiling time). Once the heater turned off, the lid was automatically pulled up by a micro gear motor. Similarly, for the aroma diffuser in Figure 4b, we placed the air-cleanser ingredient inside and covered its opening using a lid. At the end of the 22.5 min interval, the aroma diffuser turned on, and the same micro gear motor pulled up the lid. The aroma diffuser was kept running at its default settings.

Figure 4c shows the sensor setup containing all the sensors and microcontrollers listed earlier. The electrical contacts of the equipment were all soldered to ensure a minimal contact resistance and a firm connection. All of the wires were twisted to reduce cross-interference.

Figure 4d shows a 3-dimensional representation of the experimental setup drawn using SketchUp software. The water heater and aroma diffusers were turned on one at a time. Moreover, the heater in Figure 4a and the aroma diffuser in Figure 4b were continuously cleaned with distilled water to avoid contamination from previous experiments.

The experiments were carried out in an isolated and unmanned room, while the control and data acquisition and monitoring were done from a computer placed in another room. The experimental setup and the computer were connected via data and power wires taken through a tiny drilled hole in the wall. To ensure the experiment runs without error, a webcam was placed near the water heater/aroma diffuser (Figure 4d) to check that the aroma diffuser turned on successfully and the lid was pulled up properly. The webcam gave us visual feedback of the motor-pulley system operation on the screen at the top right corner of the LabVIEW frontend, as shown in Figure 4e.

When the system runs accurately, visual feedback indicates that no manual intervention is required. However, if the motor-pulley system fails for any reason, the system can be overridden to manually turn on the apparatus or control the motor manually from the LabVIEW. The manual control options are shown at the bottom right corner of the LabVIEW frontend in Figure 4e.

Figure 4e shows the frontend interface of LabVIEW. LabVIEW fully automated the entire system, comprising time control, movement of the lid, and gathering data from MATLAB and sensors, etc. The final output from LabVIEW was an Excel file created in a designated drive location.²⁰ The Excel file contained approximately 5400 rows of data from the meter (PM2.5, HCHO, TVOC, temperature, humidity, and CO₂) and sensors (PM1.0, PM2.5, PM10.0, and CO, organic gas, HCHO, humidity, temperature, and CO₂).

The measurement sequence is listed in Figure 5. LabVIEW enabled us to maintain a similar experimental sequence across all the experiments. LabVIEW also contained manual control and visual feedback to check whether the experiment ran error-free.

3.5. Experimental Troubleshooting. Sensors in real life are erroneous. While testing different sensors, the MQ series, CCS811, and GP2Y1010AU0F dust sensors were discarded since they are inaccurate and noisy. We replaced those with PMS5003 for the dust, mics5524 for the volatile organic gas, MQ7 for the CO, GM1815 for the HCHO, and MH-Z19B for the CO₂ sensing.

Initially, the sensors used in our setup did not run in synchrony. So, a sensor such as the HCHO sensor would not allow other sensors to communicate via UART communication. This had to be solved using two different Arduino boards.

The sensors, cameras, and motors require more power to operate than the computer with USB can provide. We used a 5 V and 10 A rated switched-mode power supply (SMPS) to power the system. Moreover, we used a high-speed USB hub to connect the system to the computer.

The web camera for running the OCR did not properly focus on the meter display and would appear hazy initially. Thus, two converging lenses from magnifying glasses were used to adjust the focus. Sharp and clear visuals were crucial for error-free recognition. Additionally, we updated the MATLAB code by introducing ROI, allowing the code to run OCR only to the areas marked on the screen and ignoring the background, as shown in Figure 3b.

3.6. Sensor Calibration. **3.6.1. Manual Calibration.** Before starting any experiments, we ensured that the regional PM2.5 concentration value detected by a government-certified station (data acquired from <https://aqicn.org> from the Dhaka US Consulate) located near the experimental site is above ~150 µg/m³, with an allowable tolerance of ±25 µg/m³.

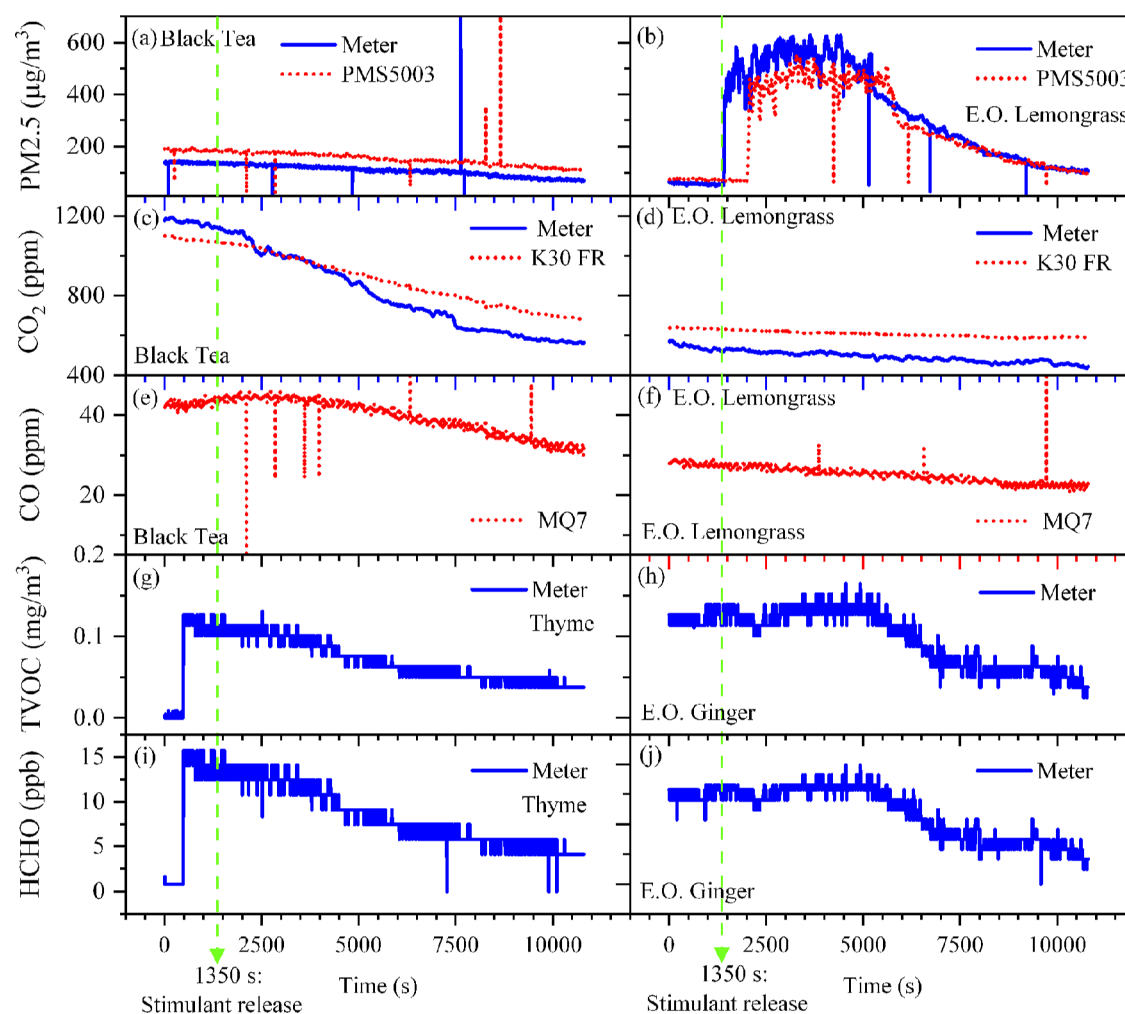


Figure 6. Measured concentration levels of boiled stimulant vapor (left column) and essential oil vapor (right column): (a,c,e) PM_{2.5}, CO₂, and CO for black tea, (b,d,f) PM_{2.5}, CO₂, and CO for lemongrass essential oils, (g,i) TVOC and HCHO for boiled thyme, and (h,j) TVOC and HCHO for ginger essential oil. All measurements were carried out up to 10,800 s, while each stimulant was introduced at 1350 s, indicated with the green dashed-line arrow.

3.6.2. Collocation Calibration. Our sensor-based measurement system was placed adjacent to a commercial AQI meter to compare the sensor data for accuracy (Figure 4d). However, a correction curve is not generated in our work as we used a comparative analysis of the final and initial AQI parameters. The raw sensor and meter data for various AQI parameters are illustrated in Figure 6 for a few representative stimulants.

The PM_{2.5} and CO₂ data recorded from both the meter and sensors closely resemble each other, confirming their accuracy. The CO quantities were measured only by the MQ7 sensor and were not referenced to any meter as our meter cannot monitor CO concentrations; nevertheless, the Keystudio MQ7 sensor consistently produced reliable and reproducible data; therefore, no reference was used. On the other hand, the TVOC and HCHO data were only recorded from the meter since some of the sensor data for these gases was erroneous.

3.6.3. Factory Calibration. **3.6.3.1. AQI Meter.** The commercial meter was kept on for 1.5 h to stabilize the data.

3.6.3.2. Keystudio MQ7 (CO). We used the analogue output of the Keystudio MQ7 CO sensor, which generates the output voltage using a 1.5 kΩ fixed resistor built into the module. The sensor continued to run for the regular preheat time of 1.5 h. A

longer suggested preheat time was avoided since it hindered continuous data acquisition.

3.6.3.3. Senseair K30 (CO₂). We used digital output data with I²C communication from a Senseair K30 FR CO₂ sensor. This sensor is well-calibrated by the factory and only requires a 1 min warm-up time, which is way less than our fixed preheat time of 1.5 h. The sensor is RoHS directive compliant.

3.6.3.4. Plantower PMS5003 (PM_{2.5}). We used the digital data output from a Plantower PMS5003 PM_{2.5} sensor. This sensor only requires a 30 s run for wake-up from the sleep mode, which was well ensured through our initialization time of 1.5 h. This sensor does not require specific user calibration. Instead, collocation calibration with a reference AQI meter and postprocessing of acquired data are suggested. In our experiments, the PM_{2.5} concentration raw data from the sensor were very similar to those from the meter. Therefore, no further modification was necessary.

3.6.4. Warm-Up and Cool-Down Time. A warm-up period of 1 h and 30 min was maintained for all measurements after a whole night of cool-down (Figure 5). Only one experiment was carried out per day to ensure similar starting conditions for the sensors.

3.6.5. Baseline Measurements. We conducted 22.5 min of baseline measurement in an undisturbed indoor environment with the sensors and the meter before the stimulants were released (Figure 5 and the left side of the green dashed line of each figure section of Figure 6). This phase confirmed uniform initial conditions for all experiments.

3.6.6. Comparative Analysis with Percentage Change. Instead of using the raw sensor and meter data, we computed the percentage change of all AQI parameter values measured over 157.5 min (the right side of the green dashed line of each figure section in Figure 6).

The percentage changes of AQI parameters are computed using the following equations:

$$\frac{\Delta Q}{Q} = \frac{Q_{\text{final}} - Q_{\text{initial}}}{Q_{\text{initial}}} \times 100\%$$

Here, Q_{initial} is the AQI parameter measured at 1350 s (highlighted with dashed green arrows in Figure 6), and Q_{final} represents the AQI parameter recorded at the last measurement point at 10,800 s. ΔQ is the difference between Q_{final} and Q_{initial} .

The relative change reduces environmental variations and calibration mismatches between sensor and meter data and almost identically provides the overall percentage change of the environment.

3.6.7. Simultaneous Experiment. The sensor- and meter-based measurements were simultaneously initiated and ended using LabVIEW; additionally, the stimulants were released at a preprogrammed time (1350 s) for all experiments with the automated Arduino-LabVIEW-controlled motor-pulley system. Because of these time-aligned simultaneous measurements, the results obtained from comparative analysis of the meter and sensors showed very similar percentage change values for all experiments.

4. RESULTS AND DISCUSSION

The vapors from diffused essential oils and boiled herbs significantly modified the AQI parameters. In this section, we present a thorough quantitative analysis of the percentage changes in each measured AQI parameter due to the introduction of stimulants over the experiment period.

Figure 6 shows the raw trends of the measured AQI parameter concentrations for some of the best ingredients for air purification observed in our experiments. The vapor of boiled black tea pushed down the PM_{2.5}, CO₂, and CO concentrations, as indicated by their corresponding linear downward graphs (Figure 6a, c, and e). The PM_{2.5} after lemongrass essential oil diffusion rather showed a wide peak, while the CO₂ and CO levels were effectively alleviated (Figure 6b, d, and f). On the other hand, the boiled thyme vapor and diffused ginger essential oil brought down the levels of TVOC and HCHO. While the thyme vapor demonstrated a steady drop of TVOC and HCHO (Figure 6g and i), the ginger essential oil effectively reduced these gases in the second half of the experiment (Figure 6h and j).

The trend of PM_{2.5} concentration change heavily depended on the base solvent (water or oil) of the ingredient. In general, the boiled herb vapor resulted in mostly linear (rise, drop, or constant) PM_{2.5} trends, as seen in the PM_{2.5} concentration trend of Figure 6a (black tea). Due to the linear changes in PM_{2.5} concentrations after the boiled herb or spice diffusion, we compared the resulting percentage changes of PM_{2.5} concentrations in a bar graph in Figure 7, where the topmost bar

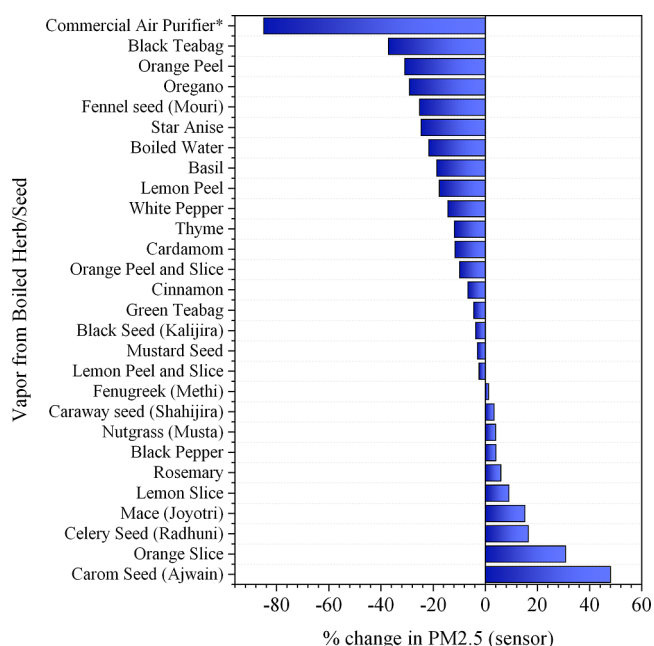


Figure 7. Percentage changes in PM_{2.5} concentration levels for various boiled herbs or seeds. Negative values indicate improved air quality.

indicates the most successful PM_{2.5} suppression. We benchmarked the results obtained from the natural air purifiers with those of a commercial air purifier. The air purifier was observed to reduce ~85% of PM_{2.5} at the cost of an undesired increase of all other AQI parameters. In contrast, the black tea vapor lessened PM_{2.5} by 37% while simultaneously maintaining a satisfactory reduction of all of the other AQI parameters. The considerable decrease in PM_{2.5} with the commercial air purifier was possible due to the high-efficiency particulate air (HEPA) filter and the ions where the latter reduced airborne germs. The ions used in the commercial air purifier can be attributed to the increase in the other gases. Besides black tea vapor, satisfactory results in PM_{2.5} reduction were also monitored from orange peel (31%), oregano (29%), fennel seed (25%), and star anise (25%) vapor.

In contrast, the essential oils generally showed peaks, as shown in the PM_{2.5} trend in Figure 6b (lemongrass essential oil). The peak in PM_{2.5} concentration upon essential oil diffusion can be explained by the fact that any oil consists of glycerol (C₃H₈O₃) and fatty acid (CH₃(CH₂)_nCOOH) as the basic units.²¹ Hence, the presence of essential oil in the air escalates the PM_{2.5} level as the oil molecules fall within the size range of this particulate matter range. Although, as stated earlier, PM (10 and 2.5) is the most significant contributor to AQI, and a high PM reading indicates poor air quality; that concept does not apply in this case. The elevation of PM_{2.5} levels due to the pervasion of essential oils does not necessarily imply degraded air quality. The oil droplets are within the PM_{2.5} size range but are not necessarily detrimental. While numerous debates exist regarding the air-cleaning properties of essential oils,²² the air-purification through the antimicrobial compounds present in most essential oils is also a widely acknowledged fact.²³

Depending on the type of ingredients, the diffused essential oils also portray different shapes of peaks in the PM_{2.5} concentration trends, such as sharp spikes and plateau peaks. They also exhibit different spread-out and diffusion rates. Oils of citrus fruits and tangy, fragrant flowers generally resulted in

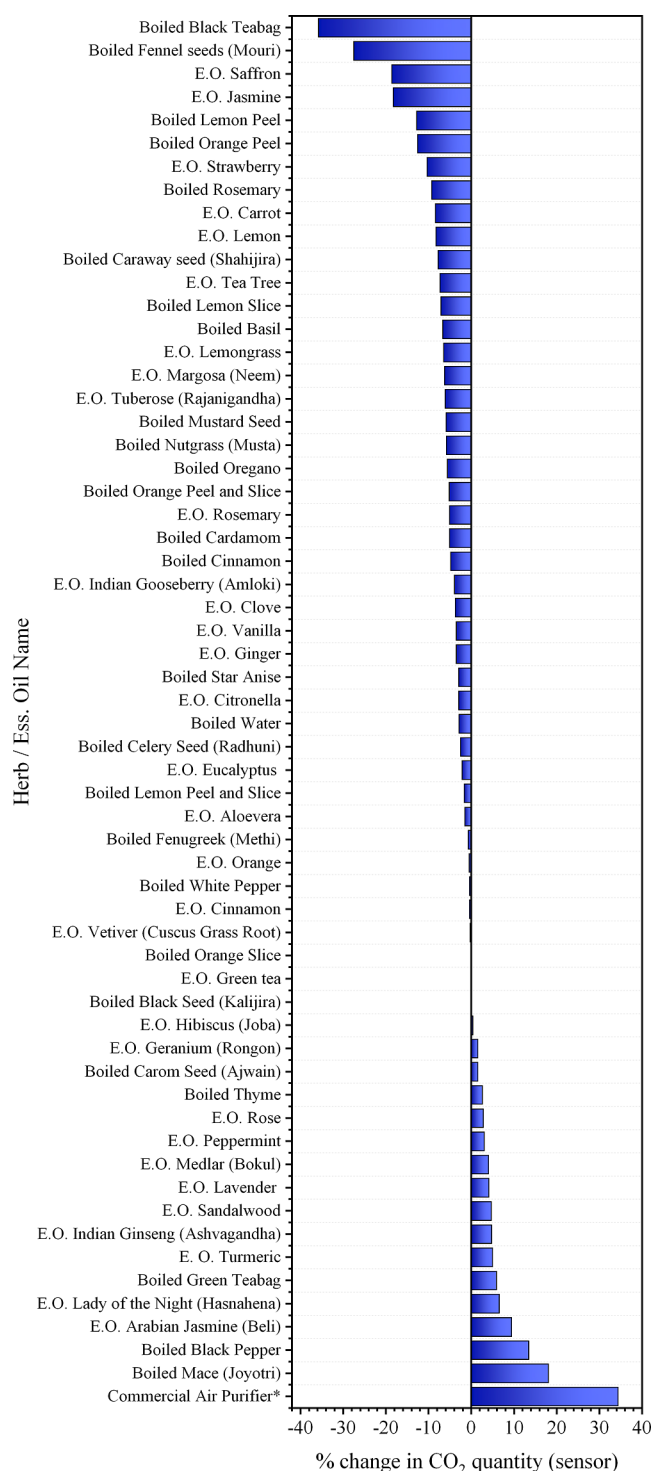


Figure 8. Percentage changes in CO₂ concentration levels due to the introduction of vapor from essential oils and boiled herbs or seeds. Negative values indicate improved air quality.

sharp peaks, followed by a steep drop. Supporting Information Note S3 lists seven types of trends of PM_{2.5} changes, with examples obtained from our experiments.

For the AQI parameter CO₂, mostly linear (rise, drop, or constant) trends were observed for most of the stimulants. The percentage changes in the CO₂ concentration due to the stimulant diffusion are illustrated in the bar graph of Figure 8. Excellent CO₂ reduction was observed from the vapors of black

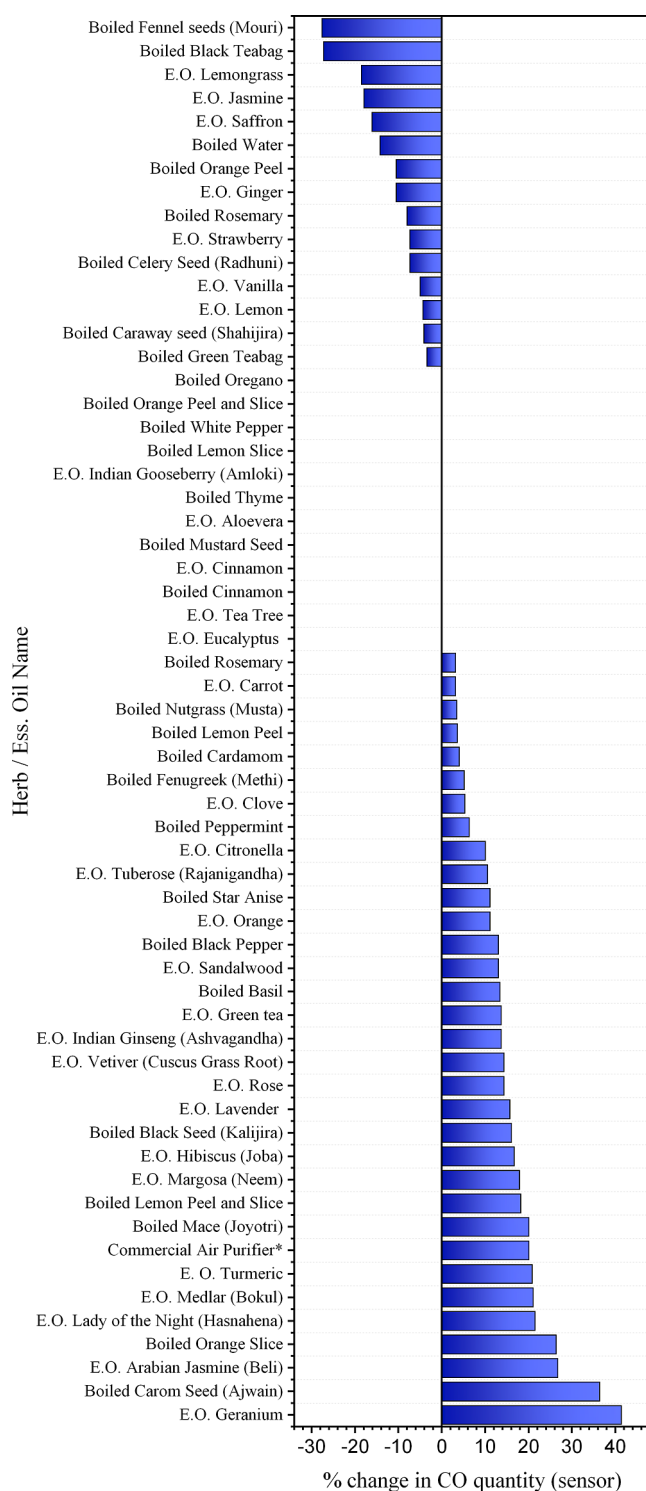


Figure 9. Percentage changes in CO concentration levels due to the introduction of vapor from essential oils and boiled herbs or seeds. Negative values indicate improved air quality.

tea (36%) and fennel seed (28%). A good number of ingredients demonstrated the reduction of the CO₂ concentration considerably, such as the boiled citrus fruit peels (13%) as well as the essential oils of saffron (19%), jasmine (18%), strawberry (10%), lemon (8%), and lemongrass (6%), which were some of the ingredients on the list. However, some ingredients deteriorated the air by increasing the CO₂ concentration, such as the essential oils of Arabian jasmine

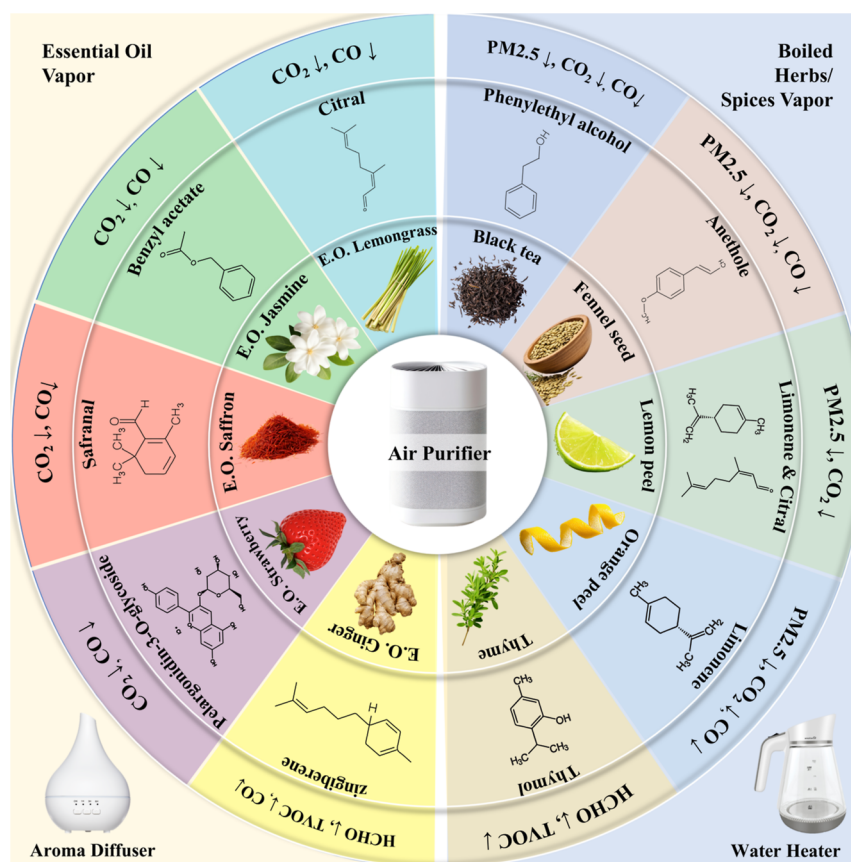


Figure 10. Natural air purification solutions obtained from our experiments. (left semicircle) Vapor diffused from essential oil and (right semicircle) vapor from boiled herbs/spices. Images of the herbs, spices, and appliances are taken from various free-domain websites without copyright restrictions.

(beli), lady of the night (hasna hena), medlar (bokul), rose, lavender, and geranium (rongon). In general, the citrus and tangy fruits (lemon and strawberry) and sharp-tangy-smelling leaves (lemongrass) lessened the CO_2 level, whereas flower-based essential oils increased the CO_2 concentration. Supporting Information Note S4 lists six types of trends with examples observed in the CO_2 changes from our experiments.

The CO levels, on the other hand, have been more or less constant in the bigger picture, with a few exceptions. For instance, boiled black tea, boiled fennel seeds, and essential oils of lemongrass and saffron lowered the CO levels more significantly than the other ingredients. A few elements escalate the CO level slightly. A large part of that list includes pebbly and block spices. The pebbly spices include black pepper, maze, and fenugreek, and block spices consist of cinnamon and aroma turmeric. Leaf extracts such as basil and green tea are included, too. Boiled orange peel and slices elevate the CO level, along with sharp-scented flowers, such as Arabian jasmine (beli) and soft-scented lavender. Surprisingly, the air purifier increases the CO level. CO_2 and CO had similar trends, though CO had a much lesser impact. Figure 9 depicts the bar graph of the percentage changes in CO concentrations.

The HCHO and TVOC data from the meter showed similar patterns in every experiment. For some essential oils, TVOC and HCHO portrayed linear trends (e.g., aloe vera essential oil), whereas some boiled fruits and herbs exhibited significant peaks (e.g., boiled lemon peel). Boiled thyme, boiled cinnamon, margosa essential oil, ginger essential oil, and a few more could lower the concentrations of TVOC and HCHO. An interesting observation is that all of these are mostly fragrant spices.

In our experiments, each stimulant was released into air in the vapor form, either by dissolving it in water or via water-diluted essential oil solvent. Hence, each stimulant encompassed the air purification quality of the vapor itself. The vapor released from boiling water purifies air in multiple ways: (i) the heat and steam in vapor kill bacteria and viruses and eliminate pollutants,²⁴ (ii) increased humidity can result in the solid-to-liquid transition of the small air particles, enhancing their diffusion and decreasing their concentration in air,²⁵ and (iii) upon cooling down of vapor, precipitation or wet deposition of small air particles reduces their concentration in air.²⁶

In the cases of essential oil stimuli, the antimicrobial, antibacterial, antiseptic, or disinfectant properties found in specific compounds aid in air purification more effectively in addition to the air-cleansing with vapor alone. For instance, limonene in lemon,²⁷ orange, and other citrus fruits is well-known for air-cleansing. Similarly, citral in lemongrass,²⁸ thymol in thyme, linalool in lavender,⁹ 1,8-cineole in eucalyptus and rosemary,²⁹ and terpinen-4-ol in tea tree³⁰ are some antimicrobial compounds familiar for air purification by neutralizing harmful substances and pollutants in air.

According to the best air-cleansing stimuli observed in our experiments, we can also attribute anethole in fennel seed,³¹ phenyl ethyl alcohol in black tea, zingiberene in ginger,³² benzyl acetate in jasmine,³³ nimbidin in margosa,³⁴ safranal and crocin in saffron,³⁵ and phenolic compounds in strawberry³⁶ as the responsible air-freshening antimicrobial agents. The summary of the most efficient air purification solutions found from our experiments is radially portrayed in Figure 10. The center of the diagram indicates that the commercial air purification machine

may aid with the natural air purifiers but is not a necessary component. On the first ring (from center), the left semicircle outlines the best essential oils while the right semicircle outlines the boiled herbs/spices. The second ring justifies the underlying reason for air purification, with the relevant antimicrobial compound for each natural air purifier. Finally, the outermost ring presents the characterized AQI parameters that were noticeably suppressed in our experiments.

5. CONCLUSION

In conclusion, air purification from natural herbs and spices is a little-known and less-explored field. Even though very few studies exist on a limited range of medicinal essential oils, the lack of experimental evidence and scientific explanations voids the credibility. In this work, we precisely characterized the air quality index parameters with an advanced, fully automated measurement setup constructed with well-calibrated sensors.

In a nutshell, we observed that a commercial air purification machine alone cannot improve indoor air quality holistically. Even though this expensive machine can remarkably suppress PM_{2.5}, it increases all other AQI, including CO₂, CO, HCHO, and TVOC. Naturally occurring herbs and spices can be used with or without the commercial air purification machine to reduce all AQI parameters comprehensively. For the best results, especially for the most sensitive group of people with severe respiratory complications, the machine can be combined with one or a few of the best stimulants observed in this work. For instance, vapor released from boiled black tea, fennel seed, or citrus fruit peel can be added to dramatically abate PM_{2.5}, CO₂, and CO quantities. On the other hand, the vapor diffused from the essential oil of lemongrass, saffron, jasmine, and strawberry can significantly assuage the CO₂ and CO concentrations. Last but not least, vapor released from boiled thyme or the essential oil of ginger can decently reduce the HCHO and TVOC concentrations.

The research outcome obtained from these studies can contribute to the healthcare field with natural medicinal and remedial alternatives for patients with dust allergies, asthma, and other respiratory complications, bypassing the risks of side effects of respiratory medications.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.5c03783>.

S1: Camera filter adjustment settings using OBS software. S2: KiCad circuit diagram of the entire system. S3: Seven types of raw data of PM_{2.5} changes represented with one example stimulant for each type. S4: Six types of raw data of CO₂ changes represented, with one example stimulant for each type. (PDF)

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Author Contributions

[†]Md.A.A. and T.T. are shared first authors. M.A.A. worked on conducting all electrical experiments, automating experiments using LabVIEW, data-filtering with MATLAB, plotting data, and assisting in manuscript preparation. T.T. worked on experimental data analysis, graph plots, and manuscript draft preparation. A.D. conducted the preliminary experiments. N.N. supervised the whole work by devising the idea, guiding the group, performing rigorous data analysis and data plotting, explaining the scientific interpretations behind the experimental observation, and contributing to manuscript writing tasks.

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Notes

We experimentally characterize the potential of natural herbs and essential oils in reducing harmful airborne pollutants to improve indoor air.

The authors declare no competing financial interest.

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